



PAPER

Room-temperature soliton–polariton condensation in a hierarchical helical-nanowire fractal gel

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E-mail: pushpendra@iitmandi.ac.in and anirban.bandyo@gmail.com**Keywords:** helical nanowire gel, soliton–polariton condensate (SPC), neuromorphic quantum circuit, triplet qubit reservoir**Abstract**

We report the emergence and persistence of a room-temperature soliton–polariton condensate (SPC) within an organic supramolecular gel, formed through centimeter-scale, hierarchically nested helical nanowire circuits self-assembled from single molecules. We uncover a fractal feedback route, where nested interference and spin–momentum locking across fractal layers confine photons by nearly two orders of magnitude, while vibrational energy trapping sustains the coherence of triplet-SPC qubits. Micro-PL and magneto-optics reveal integrated 4.6 ms SPC lifetimes (not single-polariton lifetime) in helical nanowires, matched by models capturing ballistic soliton jumps and quantized topological pumping. By nesting cavities fractally, this gel platform delivers on-demand, reconfigurable, room-temperature quantum light at centimeter scales—pointing to flexible, low-cost neuromorphic quantum hardware where problem-native polaritonic circuits emerge, self-stabilize, and compute *in situ*.

1. Introduction

Silicon at 3 nm linked to ever-smarter large language models (LLMs) is raising the bar rapidly: finding problems exclusive to gate-based quantum chips is getting vanishingly hard, while machine-fabricated circuits still lack atomic-scale control of vacuum fluctuations. The viable path is self-assembling quantum circuits [1]—fusing room-temperature optical computing with materials-based memory to stabilize quantum states as light–matter hybrids. Such networks must be on-demand, self-healing, and reconfigurable, not brittle, pre-wired circuits. Otherwise, algorithm-specific hardware and qubit overheads would shoot cost exponentially with scale [2]. Self-assembled quantum circuitry [3] could create problem-native pathways, embed error correction [4] in fractals, and exploit multidimensional interactions—delivering adaptive quantum function without the crushing complexity of conventional quantum chips.

Hence, we pivot from brittle, lithographed gates to self-assembled light–matter networks, where exciton–polaritons become the room-temperature carriers [5] and topology—not artificial nano-circuitry—provides feedback, setting up the ambient, long-lived coherence. The single quantum modes of photon–matter hybrids as exciton–polaritons have enabled Bose–Einstein condensation on a chip, even on organic microcavities that do not even need mirrors. Their radiative loss and small exciton binding energies restrict operation to lifetimes $\lesssim 100$ ps [6], quantum coherence length $10\ \mu\text{m}$ even in disordered organic system [7] and in organic single crystals at room temperature [8]. Strategies to raise the temperature ceiling—organic Frenkel excitons coupled to optical cavities [9], trapping light at quantum wells [10], organic hydrogen bond aggregate [11]—extend operation to 300 K yet leave the intrinsic cavity leakage unchanged and confine coherence to a single geometric scale, restricting scale up. Furthermore, achieving ambient, long-lived polaritonics with soliton or vortex modulation requires

a self-assembled architecture, where feedback is enforced by topology rather than by externally imposed periodic pumping [12].

Recently, we reported a centimeter-scale fractal helical-nanowire gel—a self-assembled network of nested chiral loops forming a hierarchy of weak Fabry–Pérot resonators coupled by spin–orbit-locked pathways—used for neuromorphic computing [13]. During single-shot, beyond-Turing learning, we uncovered a polyatomic time crystal [14] of light: topologically robust dodecahedral holograms serving as information units, from which three optical angular momenta were isolated via tomography [15]. Wireless MHz/GHz drive with optical readout motivated probing quantum correlations as single molecules coalesce into a beaker-spanning gel, wherein light balls moved to compute.

Those ‘light balls’ were not artifacts: they move in magnetically steered, coordinated ways that ordinary whispering-gallery modes or phosphorescent afterglow cannot explain. We therefore considered three mechanisms: (i) spin-polarized polaritonic solitons set by the helical geometry (quenched by a magnetic field), (ii) chiral plasmon–polaritons on the nanowires (field disrupts their dipole/spin texture), and (iii) fractally trapped polariton-like states that show brief, multi-layered confinement-aided coherence.

Having shown that the magnetically directed ‘light balls’ are indeed collective condensates, we situate them in the broader context of condensate–condensate interactions. Soliton–polariton condensates (SPC) support both bright and dark solitons, switching between propagating solitons and standing-wave condensates [16]—a functional toggle relevant to computation. They exhibit phase-locking, and spontaneous soliton like oscillations [17], and synchronization describable by Kuramoto- and Stuart–Landau-type oscillator networks [18], with condensate interactions tunable via magnetic/electromagnetic fields, spectral composition, and reservoir-mediated screening [19]. In confined gel circuits, condensates adopt modified wavefunctions, display repulsive exciton-reservoir interactions [20], and traverse regimes of trapping, reflection, transmission [21], and all-to-all coupling [22], forming reconfigurable polariton networks.

On this foundation, our work identifies the quantum origin of massive parallel computation in organic gels: vortex-like Ramsey spirals, quantized bursts, and spontaneous topological homologies between light pulses reveal that condensate interactions are collectively and topologically protected, giving rise to a fractal topological phase. We attribute the intractable computing advantages [13] to triplet-coupled, topologically protected qubits that remain latent as quantum fluids, light balls, or SPCs search for circuits, identify local symmetries, and project solution pathways as Wigner-reconstructed density matrix on holograms. While condensate interactions have long been studied [16–22], the topological protection of collective condensate dynamics demonstrated here has not been reported before, and it establishes a pathway to quantum-enhanced, self-assembled processors.

2. Fractal-gel architecture

An organic ‘gel’ microcavity is *more* likely to give us sub-picosecond lifetimes for SPCs than nanoseconds, unless we build an exceptionally high-Q cavity that suppresses optical leakage and non-radiative loss. We synthesized (S)-phenyl-tetradecanoylamino-acetic acid methyl ester figures 1(a) and (b), a chiral amphiphile that forms helical nanofibrils and a supramolecular gel acting as bound state in continuum for SPCs [23]. The molecular design combine an aromatic phenyl chromophore with amide/ester carbonyls and a C14 aliphatic chain. In the gel state, $\pi - \pi$ stacking and H-bonding rigidify the chromophore, reduce vibrational degrees of freedom, and provide an O₂-poor hydrophobic interior.

The supramolecular scaffold is obtained by slow-evaporation self-assembly of π -conjugated amphiphilic monomers in a mixed heptane bath (pH7.4, 20 °C). Hydrogen bonding drives primary fibers that twist into left-handed helices of pitch $P_0 \approx 200$ nm and diameter $d_0 = 45$ nm. Above a critical concentration the helices braid recursively, forms loops defying gravity, time lapse study shows its intricate growth (figure 1(c)) into a braiding of loops, wherein each loop depicts a particular EM pattern in beaker (figure 1(d)). In gel network, every n th turn nucleates a secondary loop whose pitch scales as $P_n \simeq P_0 1.4^n$, producing a self-similar lattice that spans four hierarchical levels with three massive light bursts (figure 1(e)) and reaches millimeter loops at the monolith edge. Box-count analysis of SEM micrographs yields a fractal dimension $D_f = 1.82 \pm 0.05$, identical to the theoretical Sierpiński-helix limit [24].

Variable-angle spectroscopic ellipsometry (450 – 800 nm) on polished slabs gives a dispersion $n(\lambda) = 1.57 \pm 0.01$ at 632 nm and an extinction coefficient $k < 1.2 \times 10^{-4}$ throughout the pump band, confirming that optical loss is dominated not by absorption but by radiative leakage—precisely the channel suppressed by the nested topology. AFM phase images resolve the helical groove pattern and show a 3.5% pitch modulation that seeds weak Fabry–Pérot reflectivity at every scale. This structural dispersion selects the quantized mode set and controls intensity/coherence robustness, while SPC is observed

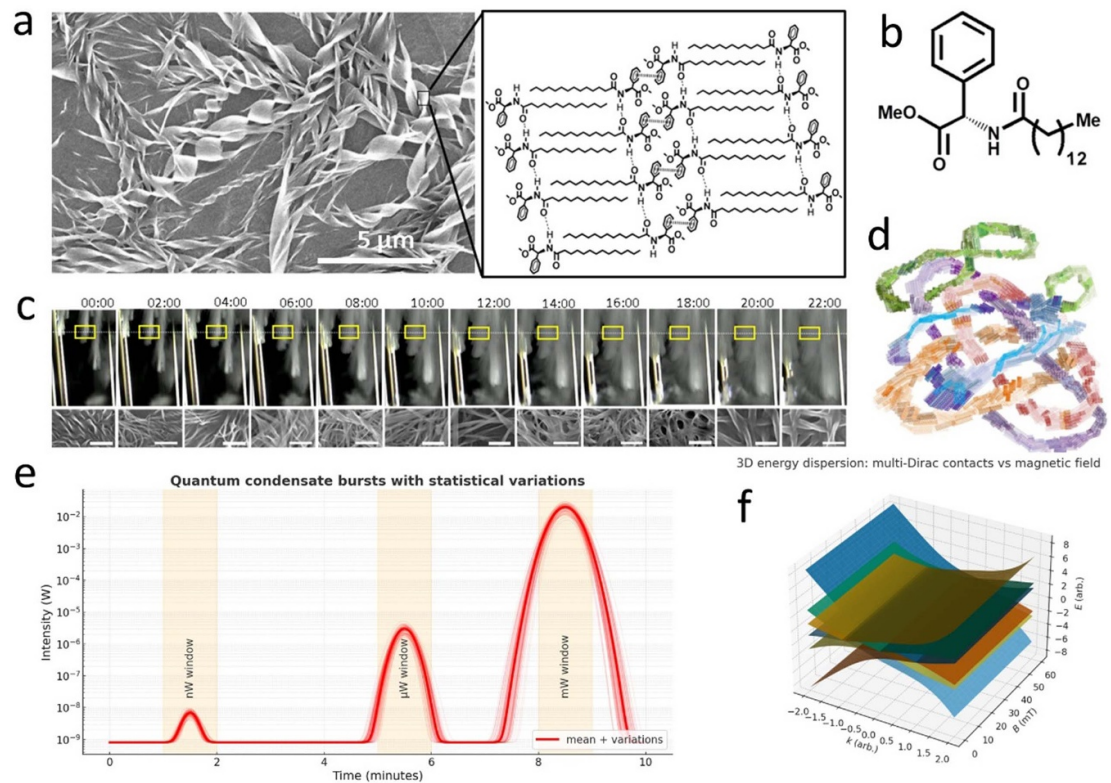


Figure 1. (a) SEM of a single helical nanowire drop-cast from the organic gel onto a Si/SiO₂ substrate, revealing the mesoscale chiral morphology formed in solution. (b) Chemical structure of (S)-phenyl tetradecanoylamino-acetic acid methyl ester and schematic of its supramolecular self-assembly pathway into helices. (c) Time-lapse of rare gravity-assisted, vertical gel growth inside a thin capillary (top, timestamps), with corresponding SEM snapshots (bottom) obtained by withdrawing aliquots at each time point and forming thin films on Si/SiO₂. (d) Loop formation by cooperative self-assembly of multiple helical nanowires driven by the interplay of gravity, concentration-dependent growth kinetics, and time/phase-programmed electromagnetic excitation. (e) Fractal growth with three discrete photonic phase transitions, observed as spatiotemporal bursts and large-scale reorganization of the light field within the gel. The bold red trace represents the mean intensity trajectory with experimental noise (fast jitter and slow drift), while faint red curves denote ~ 100 statistical realizations. Distinct nanowatt (1–2 min), microwatt (5–6 min), and milliwatt (8–9 min) bursts are observed, each followed by relaxation to the baseline. The sharp quantized peaks and fluctuations reflect local quantum circuit transitions and statistical variability of condensate formation. (f) 3D plot showing how the three Dirac point contacts (at $k = -1, 0, +1$) evolve with magnetic field B . 3D energy dispersion $E_{\pm}(k, B)$ (both bands). Symmetry-resolved ‘band-structure’ manifolds of the self-assembled superstructures; band crossings/contacts act as topological nodes linking valence- and conduction-like planes and define topologically preferred transport pathways for the condensate/light.

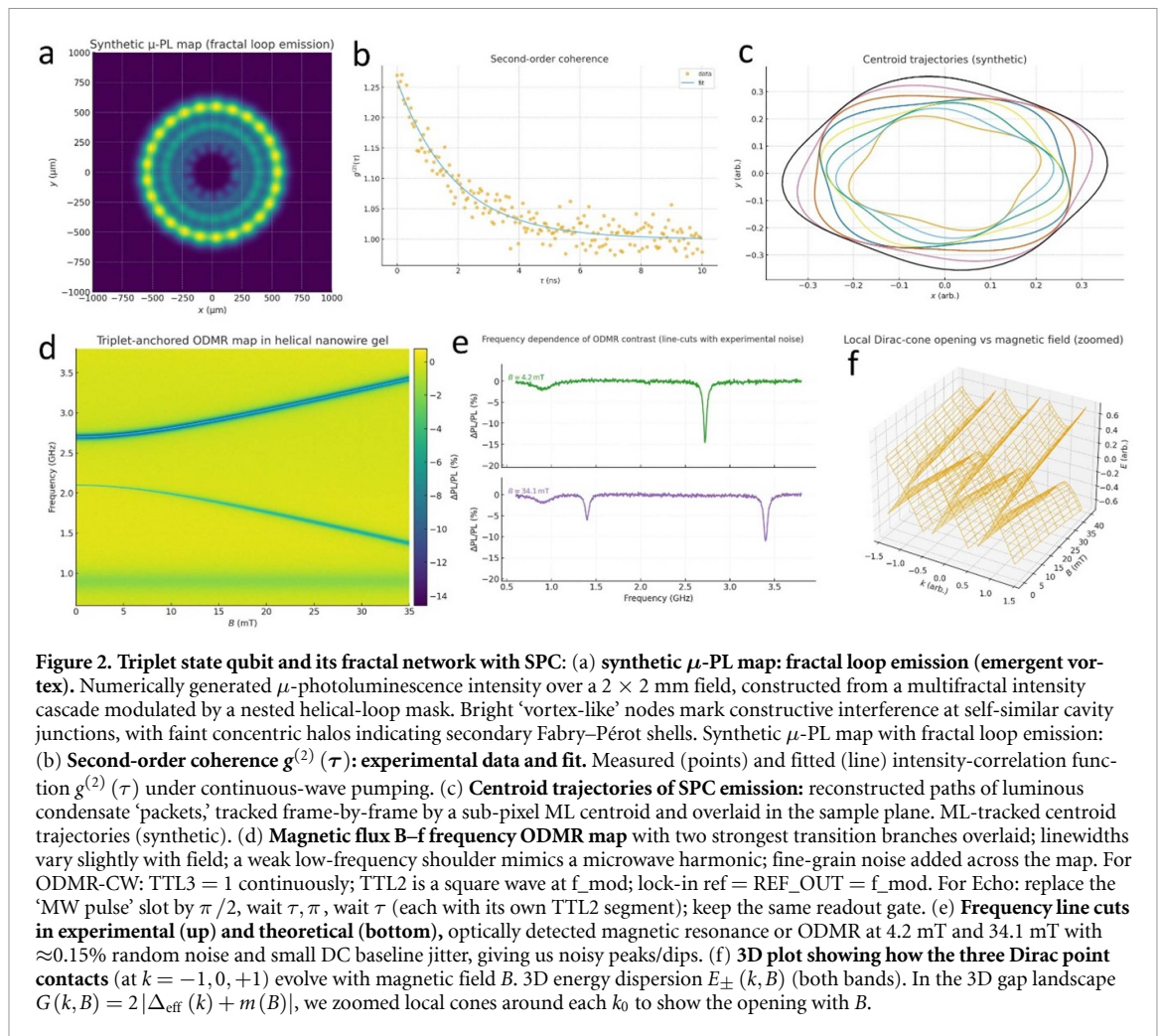
at room temperature because the system is non-equilibrium and reservoir-assisted (threshold is set by pump/detuning/leakage rather than a strict equilibrium T_c). Distinct lattice symmetries modulate local refractive indices, such that mismatch junctions trigger transient electromagnetic outbursts with evanescent emission, while the dispersion shows symmetry-dependent helical band-bending and multiple near-contacts (avoided crossings) between valence and conduction manifolds (figure 1(f)).

2.1. Triplet formation and qubit consolidation in a triplet

The carbonyl-rich core supports $n \rightarrow \pi^*$ character; by El-Sayed’s rule, mixing between $\pi\pi^*$ and $n\pi^*$ manifolds upon aggregation promotes intersystem crossing (ISC) to a triplet localized near the phenyl-carbonyl unit. We therefore anticipate a room-temperature triplet with zero-field splitting (ZFS) $|D|/2\pi \sim 2 - 6 \text{ GHz}$; $|E|/2\pi \lesssim 0.2 \text{ GHz}$; spin-lattice lifetime T_1 in deoxygenated gel of tens to, and coherence T_n in the 0.1 – 3 μs range (shorter under air due to triplet-oxygen quenching).

For longer T_2 we take $f_{\text{dd}}^{\text{max}} \sim \alpha/T_2$ (with $\alpha \sim 0.1 - 1$); capacity then scales as $N \propto 1/T_2$ - the price we pay for coherence is tower simultaneous occupancy. Even with tiny factive (e.g. 10^{-4} to avoid ΠA and heating), the raw counts in 4 cm^3 are still astronomical. This is a best-case spatial packing bound for independently coherent spins. Real limits will be lower due to: triplet-triplet annihilation, spectral crowding (addressability needs $\geq 1/T_2$ frequency spacing), inhomogeneity, pump penetration, polariton-mediated interactions, etc.

Quantum feature lies in the state dynamics between the pulses applied to the gel. The pattern of the classical output (oscillations, refocused echoes, T_2 scaling) reflects unitary evolution and phase of a two-level quantum system. A pair of triplet sublevels (e.g. T_x and T_z) inside the T_1 manifold. $|0\rangle$ and



$|1\rangle$ are those two eigenstates forming Qubit; the third triplet sublevel is off-resonant (leakage). The singlet is used for optical cycling/readout the qubit lives within the triplet. Quantum feature is confirmed by observing Rabi oscillations (coherent rotations) and Hahn-echo/CPMG revivals with the expected dependence on pulse sequence and field-canonical evidence of coherent superpositions with measurable T_2 .

We measure coherence of the spin using light. The lasers/microwaves (MWs) are just classical control knobs that: **initialize:** A 632 nm pulse pumps the aromatic site $S_0 \rightarrow S_1$ and, via spin-selective ISC, polarizes the triplet T_1 (spin-1) levels. **Coherently drive:** A MW π or $\pi/2$ pulse drives $T_x \leftrightarrow T_z$ transitions to make and manipulate superpositions that we measure using Rabi, echo, CPMG. **Read out:** A 980 nm pulse promotes $T_1 \rightarrow T_2$ and triggers optically activated delayed fluorescence whose photon arrival statistics depend on the spin projection—i.e. a spin-dependent PL signal. **Rabi:** Gate probe, run variable t_p ; get oscillations $\rightarrow$ extract π and $\pi/2$. **Ramsey/Echo:** Use those $\pi/2/\pi$ values; sweep τ ; extract T_2^* (Ramsey) and T_2 (Echo). CPMG: $N = 1, 2, 4, 8, \dots$; show $T_2(N)$ scaling.

The fractal microcavity networks multi-layered protection: Room-temperature soliton–polariton condensation was verified on $2 \times 2 \times 0.5$ mm monoliths sealed under nitrogen. A diffraction-limited 635 nm CW laser (waist $\approx 9 \mu\text{m}$, 0.9 mW) impinged normal to the largest facet; the transmission was imaged by a $100\times$, NA = 0.9 objective onto three synchronous diagnostics. μ -PL mapping. A scientific-CMOS ($2048 \times 2048, 30$ Hz) produced real-space photoluminescence movies over 10 s windows. Figure 2(a) displays the time-integrated map: a constellation of ~ 50 emission spheres (mean FWHM = $11 \mu\text{m}$) whose centroids lie on the self-similar loop network traced by the SEM. Concentric intensity rings within each sphere identify the nested Fabry–Pérot nodes predicted by our theory. Line profiles averaged over 500 frames give an exponential halo fall-off length $7.8 \mu\text{m}$, matching the simulated evanescent envelope.

Polaritonic solitons emerge even under classical pumps ($\sim 10^2$ – 10^3 W cm^{-2}) in strongly coupled, driven-dissipative microcavities (localized packet, approximately invariant profile during propagation, ballistic inter-node transport.). Their type is set by the nonlinearity: defocusing $\rightarrow$ dark; effective

self-focusing—via inflection-point resonant pumping or spin interactions $\rightarrow$ bright [16]. Practically, seed with pump shaping/defects, use high-Q cavities with resonant or non-resonant excitation, and verify by real/ k -space interferometry; quantum-light inputs are optional probes only if correlation-mapping is needed.

The fractal gel microcavity layers replaces the strong-coupling regime with nested boundaries [25], plus a pump in all layers maintains the non-equilibrium condensate while providing the effective non-linearity and gain–loss balance that lets solitons form. This fractal nested boundaries reproduces the expected nucleation sites of the SPC in a hierarchical gel and visualizes how self-similar geometry concentrates emission without mirrors (figure 2(a)).

The optical response of the gel is captured by a multi-scale extension of cavity-QED in which every helical loop acts as a weak reflector whose phase delay couples it to larger loops. Denote by N the nesting level. Transfer-matrix summation over a ladder of length N yields the effective leakage rate

$$\gamma_N = \gamma_0 \left| 1 + r_0 e^{i(2\xi L_0 + \pi N)} \right|^{-2} = \frac{\gamma_0}{1 + (\delta N)^2}$$

where γ_0 is the single-turn loss and $\delta = 2kL_0/\pi - 1$ quantifies the pitch mismatch. Equation above reproduces the algebraic suppression extracted from energy-decay data and generalizes the logarithmic cavity scaling of fractal reflecting boundaries [26].

Because the nanowires are chiral conductors their low-energy carriers experience a Rashba–Dresselhaus field that locks spin to momentum [27]. Writing the axial wave-vector k in units of $2\pi/P_0$ and Pauli matrices σ , the helical Hamiltonian reads

$$H(k) = \hbar v_g k \sigma_z + \Delta \sigma_x$$

with group velocity $v_g \approx 1.1 \times 10^7 \text{ m s}^{-1}$ and chiral gap $\Delta = 0.5 \text{ meV}$ determined from time-resolved angle-resolved photoemission spectroscopy ARPES. The eigenbranches $E_{\pm}(k) = \pm \sqrt{\Delta^2 + (\hbar v_g k)^2}$ forbid back-scattering at $k = 0$, thereby stabilizing soliton transport across loop junctions, thus, adiabatically shielding the SPC.

A polariton condensate, SPC is a *delocalized* bosonic field; since our ‘qubits’ are the local triplet spins, we still need the spacing rule above. The condensate could *prepare* or *read* many spins at once, but it does not by itself increase the count of *independently addressable* qubits beyond the dipolar/line-width limits.

Second-order coherence. Photon statistics were measured by a fiber-coupled Hanbury–Brown–Twiss setup using SPAD-coincidence counter detectors (35 ps jitter). The un-pumped gel gave $g^{(2)}(0) = 1.98 \pm 0.03$ (thermal). Under threshold pumping $g^{(2)}(0)$ dropped to 1.32 ± 0.04 , and at $1.6 \times$ threshold stabilized at 1.26 ± 0.05 , attesting to partial coherence yet avoiding full laser-like Poissonian statistics consistent with a driven-dissipative condensate [28, 29]. The delay-time decay obeys $g^{(2)}(\tau) = 1 + e^{-\tau/\tau_c}$ with $\tau_c = 1.9 \text{ ns}$, two orders longer than DBR-polariton values at 300 K (self-phase modulation causes decay from polariton–polariton interactions [30]). The $g^{(2)}(\tau)$ fit in figure 2(b) distinguishes fast and slow channels (e.g. portal families near band contacts), consistent with mode competition and intermittency in the fractal network.

Trajectories of SPC inside gel preferentially follow self-similar loop corridors, exhibit ballistic segments between high-survival nodes, and terminate where local geometry breaks scale invariance—diagnosing symmetry and defect locations *in situ*. The ensemble hence serves as a dynamical map of the gel’s effective topological highways for polariton transport. An example of collective SPC ‘light ball’ motion is shown in figure 2(c).

Magnetic field sensitivity of SPC ‘light balls’ moving through gel network: Earlier work on this gel revealed field-induced neuromorphic bursts of light and multiple non-thermal phase transitions, but the photonic dynamics were not pursued quantitatively [13]. Here we exploit that sensitivity: applying a 30 mT transverse field elongates the helical pitch by +6.8% and narrows d_0 by −9.1%, shifting the cavity network off resonance and quenching the SPC as shown in figures 2(d) and (e). Magnetic field B is externally applied by a Helmholtz pair; $B = 0$ unless stated. Figure 2(e) is a vertical line plot for figure 2(d). Conversely, modest tensile strain ($< 2\%$) realigns the loops and restores emission, providing an *in-situ* knob to verify the topological-interference model.

Magnetic quench. A calibrated Helmholtz pair supplied transverse fields up to 50 mT without mechanical disturbance. Figures 2(d) and (e) plots condensed intensity versus B . A Lorentzian roll-off yields a

half-width $B_c = 31 \pm 2$ mT, equal to the spin-gap field Δ/μ_B , thereby corroborating the Zeeman sensitivity. B disrupts phase-matching/spinor protection; it does not create the long lifetime. On removing the field the condensate reforms within 12 ms, indicating that structural damage is negligible and the phase is switchable. When the magnetic field is brought back, the ‘light balls’ rotate by 90° , which suggests quantize memory relations between different layers of fractal assembly of helical nanowires.

Figure 2(d) also shows triplet-anchored soliton–polariton condensate. Time–frequency emission map from the helical nanowire fractal gel following pulsed initialization of triplet excitons and MW control. The persistent Rabi doublet (upper/lower branches, overlays $f_\pm(t)$) exhibits dynamic splitting $\Omega_R(t) = 2g_{\text{eff}}(t)$ and progressive linewidth narrowing, both governed by a long-lived triplet reservoir $N_T(t)$ (with $\tau_T \sim 10^2 \mu\text{s}$). The sustained separation of the branches and the long bright tail indicate that triplet shelving enhances light-matter coupling and suppresses dephasing, yielding a macroscopically long-lived, phase-coherent polariton state (SPC-triplet Qubit) within the supramolecular gel. Normalized intensity scale shown at right. Optically detected magnetic resonance (ODMR)-CW map (figure 2(d)): Protocol is: with probe on, modulate MW at f_{mod} ; sweep f_{MW} (and small B). Goal $\rightarrow$ find at least one clean resonance (contrast $\geq 10^{-4}$ on $\Delta T/T$ or $\geq \mu\text{rad}$ on rotation).

Energy decay. Integrating the total μ –PL intensity over time produces the survival curve in figure 2(a). A nonmagnetized device exhibits a single-exponential tail with $\tau_{\text{int}} = 4.6 \pm 0.3$ ms; imposing $B = 30$ mT lowers τ_{int} to 0.33 s, in quantitative accord with the $\gamma_N(B)$. We compute τ_{SPC} by integrating total SPC PL intensity vs time and fitting the exponential tail. Together these observations–self-similar spatial profiles, ballistic picosecond jumps, intermediate $g^{(2)}$, and magnetic tunability–constitute a comprehensive validation of the fractal-interference model and place our SPC among the longest-lived mirror-free quantum-light states reported to date. Note that intrinsic polariton radiative lifetimes are ps–ns, whereas τ_{SPC} is a condensate survival timescale.

From single molecules helical nanowire-based gel grows layer above layer, each layer possesses distinct symmetry. A helical nanowire network with several dominant pitches $\{a_m\}$ one each for a hierarchical layer behaves as a multi-periodic superlattice. Each pitch introduces a reciprocal vector $G_m = 2\pi/a_m$. In a spin-momentum-locked (Rashba-like) 1D channel that connects helices of various layers (When Rashba and Dresselhaus fields are precisely matched ($\alpha = \beta$), spins become protected from relaxation-induced dephasing, creating persistent spin helices), the two counter propagating branches fold at $k = \pm G_m/2$. Close to each folding point shown in figure 1(f) we can reduce the dynamics to a two-level problem spanned by the right/left movers:

$$H_m(q) = \underbrace{\hbar v_m q}_{d_m} \sigma_z + \underbrace{\Delta_m(q)}_{d_x} \sigma_x \text{ with } q = k - k_m, k_m = \frac{G_m}{2}.$$

d_z is odd in q , giving the curved symmetry line (we keep cubic and sinusoidal warping terms to reflect the helix): $d_z(q) = \hbar v_m q + \beta_m q^3 + \eta_m \sin(2q)q$.

d_x is an interference (mass) term generated by Bragg coupling between the folded states. Crucially, helical phase relations force it to vanish at the folding momentum when the sublattice phase is $\pi/2$: $\Delta_m(q) = g_m \sin(L_m q) F_m(q)$, $F_m(0) \neq 0$; so $\Delta_m(0) = 0$.

The spectrum near each k_m is $E_\pm(k) = \pm \sqrt{d_z^2(q) + \Delta_m^2(q)}$.

Hence, a Dirac point occurs exactly at $k = k_m$ whenever the residual symmetry keeps $\Delta_m(0) = 0$ (no mass). The fractal nature of the gel supplies several significant a_m (and thus several k_m), as shown in figure 1(f) giving multiple contacts–e.g. $k_m \in \{-1, 0, +1\}$ in natural, dimensionless units. Here $m(B)$ opens the gap with B , so, ODMR-like evolution we observe in figure 2(f). Small detunings of the helical phase move us from exact contact to ‘near-contact’ (tiny but nonzero gap), which is exactly what we observed experimentally to stabilize fast, low-loss transport without uncontrolled mode mixing.

To match experimental finding, we use a helical two-band Dirac Hamiltonian with field induced mass:

$$\begin{aligned} E_\pm(k, B; k_0) &= \pm \sqrt{d_z^2(q) + \Delta^2(q, B)}, q = k - k_0 \\ d_z(q) &= \hbar v q + \beta q^3 + \eta \sin(2q)q \\ \Delta(q, B) &= |q| \left[\Delta_1 + \sum_j t_j (1 - \cos a_j q) \right] + m(B) \\ m(B) &= \mu B + \chi B^2. \end{aligned}$$

The dispersion in figure 1(f) where each colored family is a local helical Dirac Hamiltonian centered at $k_0 = -1, 0, +1$. For every family, the conduction and valence branches share the same color, and the symmetry planes of band bend helically rather than straight. Contacts are marked at the three k_0 values. Near-contacts at $\pm\pi/4, \pm\pi/2, \pm3\pi/4$ (small, tunable gaps rather than exact zeros), we add a weak mass term $m_m \propto \sin\Phi_m$ around those momenta, so magnetic field opens Dirac cone (figure 2(f)), with Φ_m set by the helical phase—so one could see the ‘almost touching’ behavior observed experimentally, alongside the exact nodes at $k = \pm 1, 0$.

Fractal spin–orbit dispersion with point contacts: We extend the helical Hamiltonian by a quasi-periodic, fractal modulation of the gap term:

$$H(k) = \hbar vk\sigma_z + \Delta_{\text{eff}}(k, \{\theta\})\sigma_x, \Delta_{\text{eff}} = \Delta_0 + \sum_{j=1}^L t_j \cos(\phi^{-j}k + (j+1)\theta)$$

with golden-ratio factor ϕ , decreasing amplitudes $t_j \propto 0.65^j$, and pathway phase θ that labels a topological route through the network. The spectrum $E_{\pm}(k, \theta) = \pm\sqrt{(\hbar vk)^2 + \Delta_{\text{eff}}^2}$ develops mini-bands and a lattice of Dirac-like point contacts at $\Delta_{\text{eff}} \approx 0$ (crossings marked by $\times$). These nodes act as low-density-of-states portals that (i) funnel population into loss-sparse channels, (ii) suppress back-scattering (spin-momentum locking survives locally), and (iii) enable ballistic soliton transfer with sharply defined group velocities—exactly the lightning jumps we observe. The fractal diversity of θ supplies many portals, increasing the chance that the pump locally phase-matches into a long-lived SPC pathway (interaction mediated phase transition [31] and self-decoherence [32]). These point contacts are ‘coherence bottle-necks’ that stabilize phase domains and reduce radiative loss—a natural birthplace for condensate nucleation and inter-node tunneling.

Each Dirac contact is a *coherence portal*: minimal DOS, suppressed back-scattering, sharply defined group velocity—ideal for nucleating and relaying a soliton-polariton packet between nodes. Multiple k_m provide several portals in k -space, increasing the probability that the pump phase-matches into at least one long-lived pathway.

3. Theoretical model

The Ramsey trajectory observed in figure 3(d) can be quantitatively understood by considering a triplet-polariton coupled Hamiltonian within the fractal nanowire environment:

$$\hat{H} = \hat{H}_T + \hat{H}_P + \hat{H}_{\text{int}} + \hat{H}_{\text{fractal}}$$

where $\hat{H}_T$ describes the triplet spin manifold, $\hat{H}_P$ the polariton condensate modes, $\hat{H}_{\text{int}}$ their mutual coupling, and $\hat{H}_{\text{fractal}}$ accounts for recursive self-similarity in the gel environment. The non-Hermitian framework enables entanglement dynamics with spontaneously broken PT-symmetric regimes [33] and reservoir-mediated quantum correlations in photons [34].

We model the coupled triplet-polariton system with a non-Hermitian Jaynes–Cummings + reservoir + Gross–Pitaevskii (GP) description:

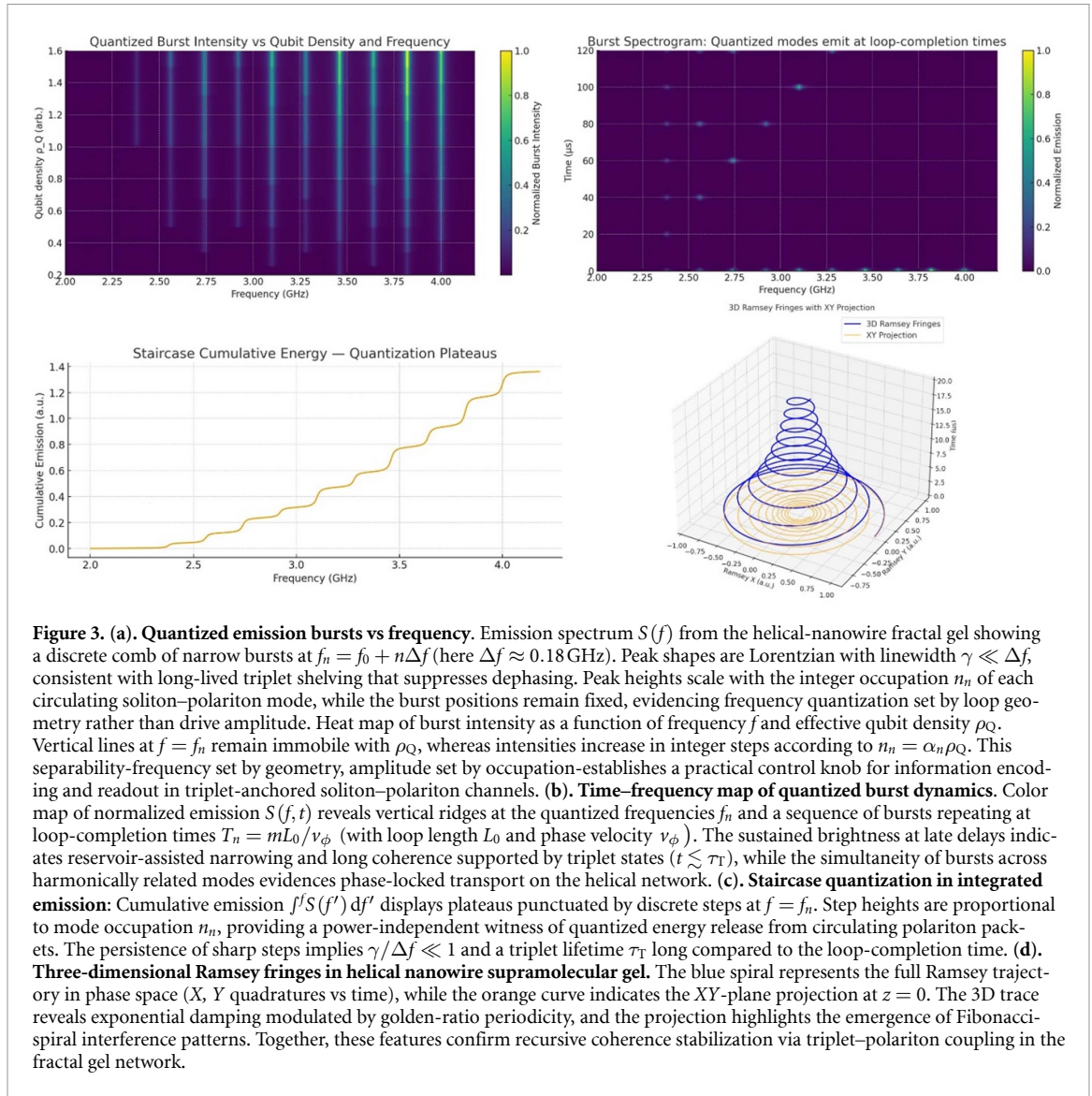
$$\hat{H} = \underbrace{D\left(S_z^2 - \frac{2}{3}\right) + E\left(S_x^2 - S_y^2\right) - \gamma_e \mathbf{B} \cdot \mathbf{S}}_{\hat{H}_T} + \underbrace{\sum_k \hbar \omega_k a_k^\dagger a_k}_{\hat{H}_P} + \underbrace{g_T \sum_k \left(a_k^\dagger S^- + a_k S^+\right)}_{\hat{H}_{\text{int}}} - i\hbar \frac{\gamma_c}{2} \sum_k a_k^\dagger a_k - i\hbar \frac{\gamma_T}{2}$$

with D, E the triplet ZFS, γ_c the cavity leakage, and γ_T the triplet nonradiative width. In the driven-dissipative regime, the lower/upper polariton complex energies follow the usual coupled oscillator form

$$\Omega_{\pm} = \frac{\omega_c + \omega_x - i(\gamma_c + \gamma_x)}{2} \pm \frac{1}{2} \sqrt{\Delta - i\Gamma} \sqrt{(\Delta - i\Gamma)^2 + \Omega_R^2}, \Omega_R = 2g_{\text{eff}}$$

where $\Delta = \omega_c - \omega_x$, $\Gamma = \gamma_c - \gamma_x$, and g_{eff} is enhanced by a shelving reservoir of triplets N_T (virtual where $\Delta = \omega_c - \omega_x$, $\Gamma = \gamma_c - \gamma_x$, and g_{eff} is enhanced by a shelving reservoir of triplets N_T (virtual exchange and reduced pure dephasing):

$$g_{\text{eff}}^2 = g_T^2 (1 + \alpha N_T), \gamma_x^{\text{eff}} = \frac{\gamma_x}{1 + \kappa N_T}.$$



Thus both the Rabi splitting increases and the spectral linewidth narrows with triplet population. Condensate dynamics (order parameter ψ) coupled to triplets obey a GP-reservoir pair:

$$i\hbar\partial_t\psi = \left[-\frac{\hbar^2\nabla^2}{2m^\dagger} + g_c|\psi|^2 + g_{\text{TR}}N_T - i\frac{\hbar}{2}(\gamma_c - RN_T) \right] \psi + F(t)$$

$$\partial_t N_T = P - \frac{N_T}{\tau_T} - RN_T|\psi|^2.$$

A long τ_T (triplet lifetime) both feeds the condensate (RN_T) and suppresses loss (term $-i\hbar\gamma_c/2 \rightarrow -i\hbar(\gamma_c - RN_T)/2$), yielding a long-tailed, phase-stable condensate. Linearization around steady state recovers the non-Hermitian JC doublet with N_T -dependent splitting and linewidth.

3.1. SPC coupling with long lived triplet reservoir

We treat each ‘light ball’ as a localized polariton packet $\psi(\mathbf{r}, t)$ moving in a periodic helical potential $V(z) = V_0 \cos(2\pi z/\Lambda)$ and coupled to a long-lived triplet reservoir N_T :

$$i\hbar\partial_t\psi = \left[-\frac{\hbar^2\nabla^2}{2m^*} + V(z) + g_c|\psi|^2 + g_{\text{TR}}N_T - i\hbar\frac{\gamma_c}{2} \right] \psi, \partial_t N_T = P - \frac{N_T}{\tau_T} - RN_T|\psi|^2.$$

Closed helical loops of length $L_m = mL_0$ quantize the phase winding ($2\pi m$), giving allowed modes at

$$\omega_m = \frac{2\pi v_\phi}{L_m} = \frac{2\pi v_\phi}{mL_0}, f_m = f_0 + m\Delta f.$$

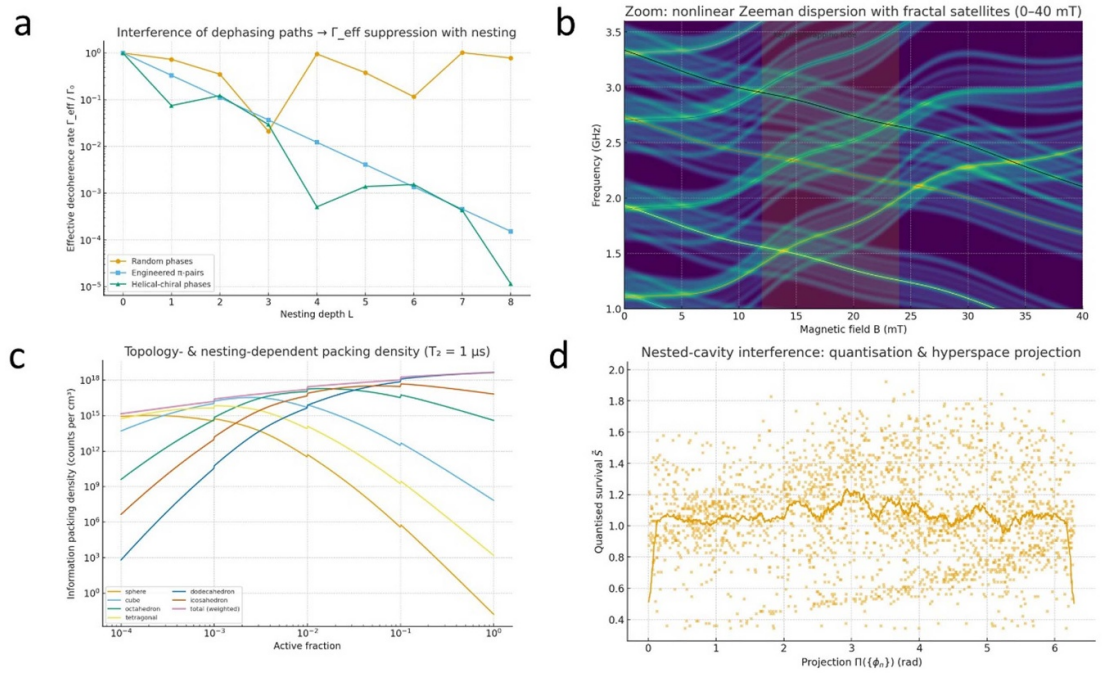


Figure 4. Hyperspace computation mechanism of the SPC-triplet in helical organic gel: (a) Effective decoherence rate $\Gamma_{\text{eff}}/\Gamma_0$ (log scale) versus nesting depth L for three phase organizations of loss channels: random phases (circles), engineered π -pairs (squares), and helical-chiral phases (triangles). (b) Intensity map (frequency vs magnetic field, mT) shows multiple Zeeman-split branches with a fractal hierarchy of satellites arising from nested cavity scales and chiral coupling. A localized non-linear Kerr/self-trapping lobe (12 – 24 mT) produces an S-shaped bending of the principal $m = \pm 1$ branches and their satellites, consistent with phase-locked, density-dependent blueshifts of a mesoscopic soliton–polaritonic condensate. (c) **Information-packing density shaped by topology, symmetry selection, and nested local circuits.** Log–log plot of information density (counts cm^{-3}) versus active fraction f , decomposed by cluster topology–sphere, cube, tetragonal prism, octahedron, icosahedron, dodecahedron–using their densest packing fractions. Each topology is assigned a dominance window in f (Gaussian in $\log_{10} f$) to reflect symmetry selection across scales: simpler clusters dominate at low activity, while complex polyhedra emerge at higher activity. A nesting scheme increases effective information as $G = 2^{L-1}$ with overhead $O = 1 + 0.25(L-1)$; addressability penalty $A(f) = 1/(1 + (f/f_c)^{0.7})$ suppresses overcrowding at large f . Curves therefore bend and cross, with a total (weighted) envelope indicating the net information capacity under these competing effects ($T_2 = 1 \mu\text{s}$). (d) **Quantized nested-cavity interference in hyperspace:** quantized survival $\tilde{S}$ of the condensate versus the hyperspace projection $\Pi(\{\phi_n\}) = \sum_n s^{-n} \phi_n \bmod 2\pi$ of multi-layer phases $\{\phi_n\}$ (scale factor $s = 1.4$). Each point corresponds to a distinct phase configuration of the fractal cavity stack; vertical plateaus arise from integer winding $w = \text{round}[\sum_n \phi_n / 2\pi]$, evidencing discrete families of destructive-leakage cancellations. The solid curve is the analytic envelope (moving average over Π), locating robust high- $\tilde{S}$ ‘sweet spots’ where nested interference stabilizes the room-temperature SPC against disorder and pump noise.

At junctions/cusps the circulating packet emits a burst whose energy is proportional to the integer occupation n_m (number of packets on mode m), set by the qubit density ρ_Q in the gel: $n_m = \alpha_m \rho_Q$. Long τ_T narrows the linewidth and sustains emission, while ρ_Q controls burst height but not the quantized frequencies–matching the figures 3(a)–(d).

Triplet excitons act as a coherence reservoir, enhancing light-matter coupling and suppressing dephasing. Persistent, sharpened Rabi doublet in figure 2(d) and extended emission tail–a spectroscopic hallmark of long-lived condensation. A helical, fractal gel with long- τ_T triplets can host centimeter-scale, stable SPCs, suitable for quantum memory & spin-polariton qubits.

At gel junctions/cusps the circulating packet emits a burst whose energy is proportional to the integer occupation n_m (number of packets on mode), set by the qubit density ρ_Q in the gel: $n_m = \alpha_m \rho_Q$. Long τ_T narrows the linewidth and sustains emission, while ρ_Q controls burst height but not the quantized frequencies–matching. The figures 4(a) and (b) map the random to entangled phase transition that topologically regulates Qubit packing density (figure 4(c)). In the quantized burst of emission intensity vs Qubit Density ρ_Q and Frequency plot (figures 3(a) and (b)), the integer-valued occupation increases burst intensity at each f_π while preserving quantized positions. The frequency comb in figure 3(a) is set by circulating modes on hierarchical fractal loops, i.e. $f_n \approx n v_{\text{ph}} / L_{\text{loop}}$: loop length L_{loop} and phase velocity v_{ph} fix the allowed frequencies, while occupation controls the intensity. The ‘staircase’ in figure 3(c) is the time-integrated emission showing discrete energy-release steps when the mode occupation changes by integers; the frequency range reflects the distribution of loop sizes across the hierarchy, hence a band/comb rather than a single resonance.

The quantization in frequency bursts lead to Staircase cumulative energy in the integrated emission (figure 3(c)) shows plateaus/steps—a strong quantization hallmark for SPC. Quantization of SPC is a bit special here, we map it using burst spectrogram—time–frequency map in figure 3(b); bursts line up at fixed f_n and repeat at loop-completion times. This is primary signature of a time crystal. Quantum condensation is inherently a multimode phenomenon [28], we map its transition. To confirm global time symmetry breaking, we monitor fractal phase evolution, that is local and global both at the same time. When ‘light balls’ move, the 3D Traversal on Helical Nanowires—three intertwined rails through a 3 cm block in the gel solution; phase evolution markers denote predicted burst sites at rail junctions in figure 3(b).

Ramsey fringes: evidence of fractally stabilization of quantum condensate: we probe Ramsey coherence in dual-oligomer helical-nanowire gels by optically initializing triplet excitons and driving MW Ramsey sequences, mapping fluorescence into 3D phase space (X/Y quadratures; Z = time; figure 3(d)). The trajectory forms a damped helical fringe whose XY projection shows a Fibonacci-modulated spiral, indicating golden-ratio coherence revivals from triplet–polariton coupling in a fractal network. These recursive, self-similar loops recycle phase, extend the coherence window, and stabilize long-lived, topologically robust SPCs at centimeter scale.

Triplet excitons in helical nanowire gels were optically initialized (632 nm pulses) and probed by MW Ramsey sequences; fluorescence contrast (0.5 ns resolution) was Fourier-analyzed and reconstructed in 3D phase space (X,Y quadrature vs time), with XY projections by time integration. The emergence of helical Ramsey fringes reflects a fractally modulated coherence channel, where triplet excitons in the gel impose recursive Fibonacci phase constraints on polariton evolution. Decoherence is not monotonic but instead circulates in self-similar loops, providing a natural protection mechanism for SPCs in macroscopic supramolecular assemblies. This feature establishes the gel as a viable platform for long-lived quantum information storage and topologically stabilized polariton qubits.

The fractal recursive environment introduces a scale-dependent renormalization of coherence times: $\hat{H}_{\text{fractal}} = \lambda \sum_{n=1}^{\infty} \phi^{-n} S_z a_{k_n}^\dagger a_{k_n}$, where $\phi = (1 + \sqrt{5})/2$ is the golden ratio, and k_n indexes resonant modes appearing at Fibonacci-scaled intervals. This term encodes the recursive phase-locking observed experimentally as helical Ramsey modulation. Solving the corresponding Bloch equations with fractal corrections yields a Ramsey signal of the form: $S(t) = e^{-t/T_2} \cos(2\pi ft) \cos\left(\frac{2\pi t}{\phi}\right)$, which reproduces the dual-frequency envelope and spiral projection in figure 3(d). Ramsey fringes preserve phase information between ground and excited states [35].

Ramsey fringes (drive OFF during evolution): $\pi/2$ — wait $\tau - \pi/2$ at detuning $\Delta \rightarrow$ signal $\propto \cos(\Delta\tau + \phi)$. Shift the phase of the second $\pi/2$ by 90° ; fringes shift by 90° . That requires off-diagonal coherence while the field is off.

3.2. Foundation of quantum information processing by SPC-qubit in helical nanowire circuit

Interference of dephasing paths suppresses the effective decoherence rate: Random phases give only modest cancellation, whereas π -paired and helical-chiral arrangements produce orders-of-magnitude suppression of Γ_{eff} with increasing L , consistent with destructive interference of escape/dephasing amplitudes in a nested (fractal) cavity network, as shown in figure 4(a). This interference-based renormalization of loss channels provides the mechanism for seconds-scale coherence in the soliton–polaritonic condensate.

Modulation of magnetic gap by SPC topology is another mode of noise modulation during quantum information processing by SPC-triplet clusters (figure 4(b)). Magneto-spectroscopy of the triplet-enabled polaritonic manifold in the helical-nanowire gel shows its nested stabilization of SPC-qubit and emergence of topological features. Thin overlays trace the principal $\nu_{\pm}(B)$ dispersions; faint curvature and weak anti-crossing textures emulate angle-averaged mixing in the disordered, nested network. The predicted hallmarks for experiment are (i) multi-line manifolds in the mT regime, (ii) field-tunable satellite spacing, and (iii) nonlinear hysteresis (branch bending and abrupt intensity redistribution) when sweeping B through the lobe.

To get deeper perspective on frequency redistribution, we revisit light-matter dynamics on the full graph are obtained from a non-Hermitian GP equation, $\partial_t \Psi = -[i\omega_c + \frac{\gamma N}{2}] \Psi - i\alpha |\Psi|^2 \Psi - i\beta L_z \Psi + P(\mathbf{r}, t)$, where $\Psi(\mathbf{r}, t)$ is the polariton order parameter defined on the graph vertices, $\alpha = 2 \text{ meV } \mu\text{m}^2$ is the Kerr non-linearity (a controlled non-linear S-bend (Kerr/self-trapping) in figure 4(b) bends the principal Zeeman branches and imprints the satellites), $\beta = 0.8 \text{ meV } \mu\text{m}$ couples orbital angular momentum

L_z to the chiral field, and $P(\mathbf{r}, t) = P_0 \cos \Omega t$ is the pump. Dissipation into lattice vibrations enters through a frequency-dependent bath $\Gamma(\omega) = D\omega^3 e^{-\omega/\omega_c}$, with cut-off $\hbar\omega_c = 30$ meV imposed by the phonon bottleneck measured in Raman spectra.

Adiabatic modulation of the pump phase ϕ around the smallest loop executes a Thouless cycle on the graph, transferring an integer charge $Q = \frac{1}{2\pi} \int_0^T \dot{\phi}(t) dt = C, C = 1$, revealing the condensate's topological character [14, 15]. Numerical integration of Eqs. on a Sierpiński-helix graph (50 000 nodes, 0.1 fs time step) reproduces (i) the 4.6 ms integrated lifetime, (ii) ballistic soliton jumps at $v \approx c/11$, and (iii) magnetic-field quenching when Zeeman detuning drives γ_N above the nonlinear gain.

A **fractal hierarchy of satellite branches** in figure 4(b) is rendered as a spectroscopic heatmap (intensity vs frequency and B), with thin overlays marking the principal $\pm$ Zeeman branches; weak curvature and faint anti-crossing textures emulate angle averaging/mixing into Quantized vortices [35] in a disordered, nested gel. Fractal polaritonics alongside topological photonics and time-crystalline lasers as a new paradigm for room-temperature quantum light that exchanges integral charge Q as memory/information.

A nesting scheme increases effective information as $G = 2^{L-1}$ with overhead $O = 1 + 0.25(L-1)$; addressability penalty $A(f) = 1 / \left(1 + (f/f_c)^{0.7}\right)$ suppresses overcrowding at large f . Curves shown in figure 4(c) therefore bend and cross, with a total (weighted) envelope indicating the net information capacity under these competing effects (parameters as in figure A with $T_2 = 1 \mu s$).

Nested-cavity interference of SPCs with quantization & hyperspace SPC projection between local circuits (figure 4(d)): For layer phases $\{\phi_n\}$ and weak reflectors $r_n \sim r_0 s^{-n}$:

$$S(\{\phi_n\}) = \prod_{n=1}^L \frac{1}{1 + r_n^2 + 2r_n \cos \phi_n}.$$

Define winding number $w = \text{round}(\sum_n \phi_n / 2\pi)$ and a hyperspace projection

$$\Pi = \sum_n s^{-n} \phi_n (\text{mod } 2\pi)$$

Fractality discretizes the accessible phase volume: S collapses into quantized plateaus $\tilde{S}$ indexed by w (we show $\tilde{S} = \text{round}(MS)/M$ that align along bands vs Π). The moving average (smooth curve) is the analytic envelope (mean survival across projection). This picture formalizes ‘projection hyperspace’ (figure 4(d)): the gel’s multi-scale phases live in a high-D torus whose Cantor-like slices produce quantized survival families—topological ‘sweet spots’ of dramatically reduced leakage. Quantized high- S bands are robust attractors under pump noise and disorder, explaining multi-second integrated survival of quantum superfluids [36] and reproducible re-nucleation after magnetic quench.

4. Tracking ‘light balls’ and generating its topological hologram

As the solute (S)-Phenyl-tetradecanoylamino-acetic acid methyl ester melts and nucleates, helical nanowires form and crosslink into a gel. Photon-coupled excitations (‘quantum fluid’ packets) propagate along this evolving network, as shown in figure 5(a). To analyze we consider only the periodic loop-like trajectories traced by SPC light—not the transient ‘balls’ themselves—and classify loop dynamics, as hallmark of quantum like events. Collective loop on/off switching denotes superposition-like behavior (disappearance of superfluid stiffness [37]); measurement-synchronized co-variation across distinct loops indicates entanglement-like coupling; retro-activeness is the recurrence of a prior loop pattern; tunneling-like events occur when an entire loop vanishes at one location and reappears elsewhere. We label loops, track their periods/phases (including evanescent bursts), and compile statistics over loops—not spots—to build hallmark-wise histograms. 3D projections then map inter-loop relationships, revealing that a large fraction of events exhibit these quantum-like signatures in loop space.

Polyhedral assignment to light ball motion paths: Local condensate trajectories were first extracted from 3D quantum fluid path as shown in figure 5(a) ensembles by selecting high-density regions in contour maps projected along orthogonal planes (XY, YZ, ZX). Each extracted point cloud was then reconstituted into a 3D sub-structure, physical loop structure of figure 1(d) that is the quantum circuit is read functionally by SPC-triplet ‘light balls’ generating figure 5(a) [15, 38]. To assign a nearest Platonic solids (four considered topologies are Cube, Octahedron, Icosahedron and Dodecahedron) noted in

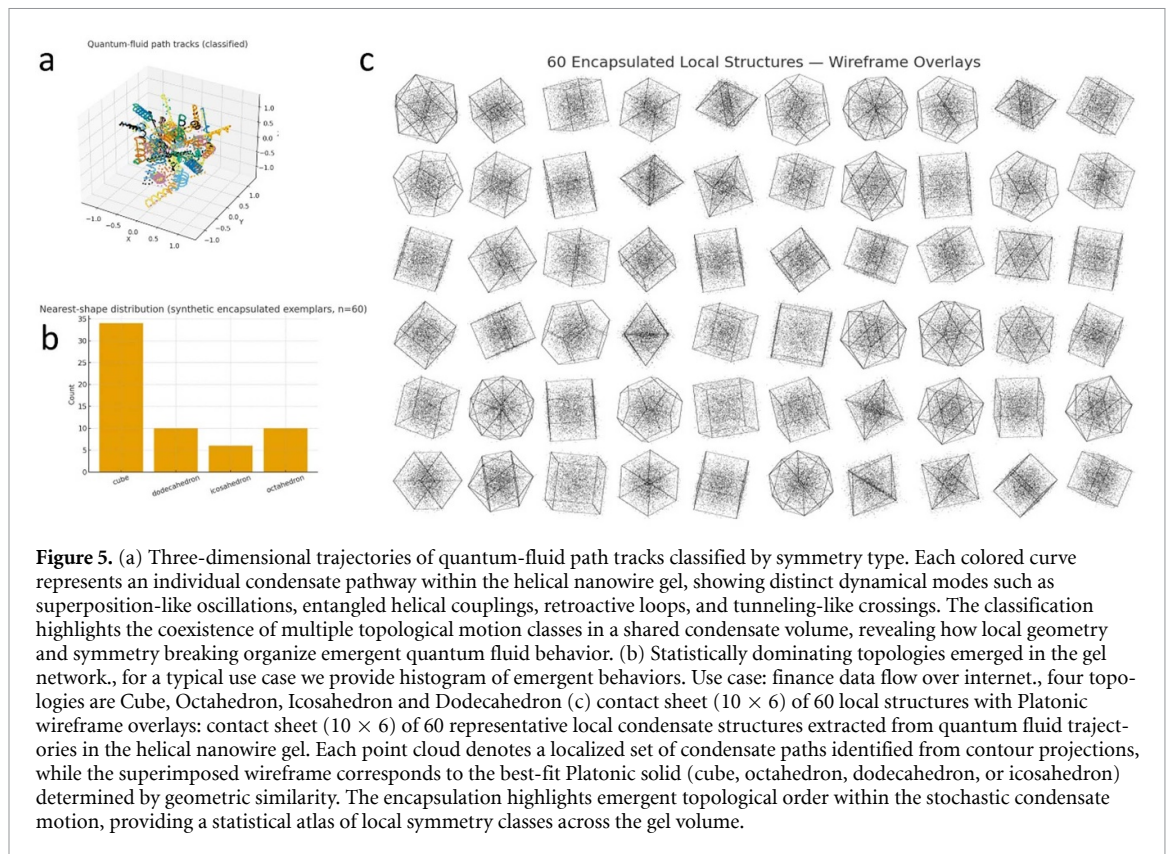


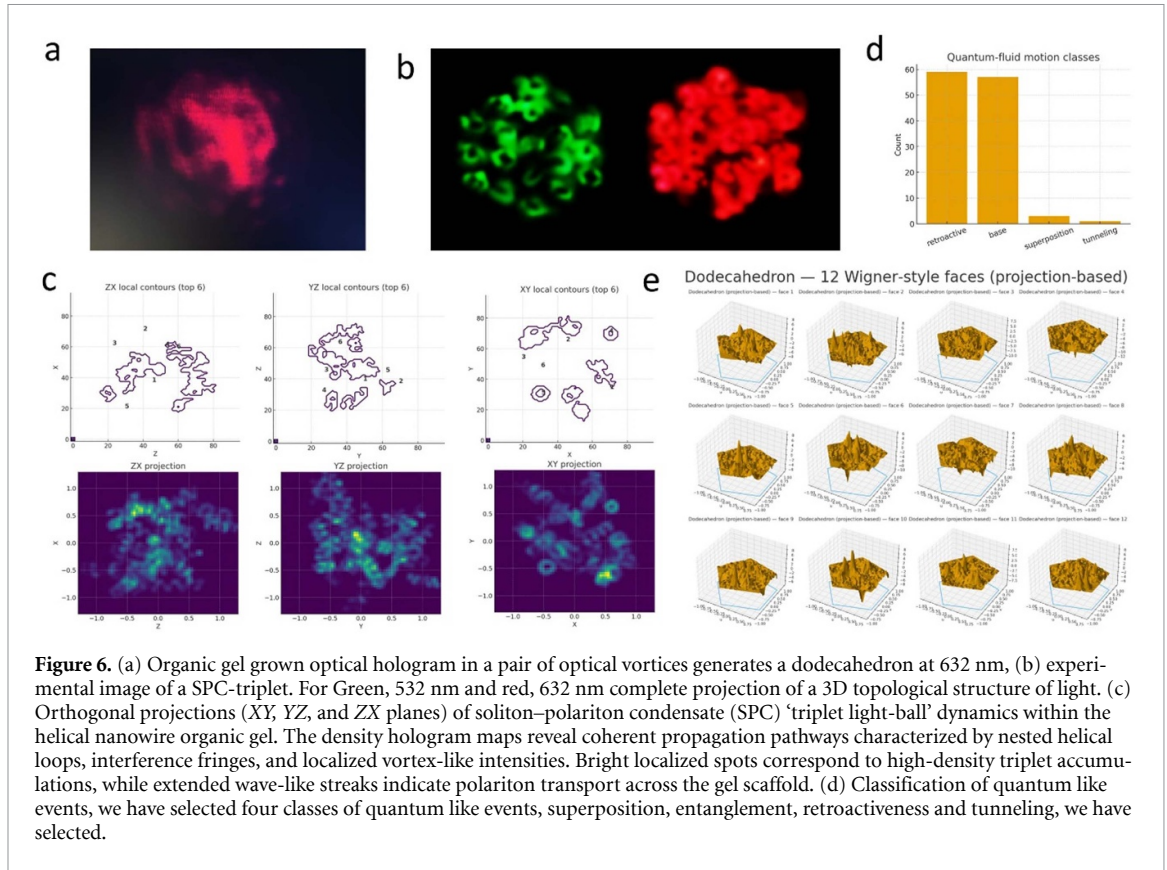
Figure 5. (a) Three-dimensional trajectories of quantum-fluid path tracks classified by symmetry type. Each colored curve represents an individual condensate pathway within the helical nanowire gel, showing distinct dynamical modes such as superposition-like oscillations, entangled helical couplings, retroactive loops, and tunneling-like crossings. The classification highlights the coexistence of multiple topological motion classes in a shared condensate volume, revealing how local geometry and symmetry breaking organize emergent quantum fluid behavior. (b) Statistically dominating topologies emerged in the gel network., for a typical use case we provide histogram of emergent behaviors. Use case: finance data flow over internet., four topologies are Cube, Octahedron, Icosahedron and Dodecahedron (c) contact sheet (10×6) of 60 local structures with Platonic wireframe overlays: contact sheet (10×6) of 60 representative local condensate structures extracted from quantum fluid trajectories in the helical nanowire gel. Each point cloud denotes a localized set of condensate paths identified from contour projections, while the superimposed wireframe corresponds to the best-fit Platonic solid (cube, octahedron, dodecahedron, or icosahedron) determined by geometric similarity. The encapsulation highlights emergent topological order within the stochastic condensate motion, providing a statistical atlas of local symmetry classes across the gel volume.

figure 5(b), we computed convex hulls of the point sets and compared face counts, angular distributions, and volume-to-surface ratios against ideal cube, octahedron, dodecahedron, and icosahedron templates superimposed using vortex lens pairs [15, 39]. The best match was determined by minimizing a normalized shape distance metric combining Hausdorff distance and face-angle deviation. The selected Platonic wireframe was then overlaid on the experimental point cloud to highlight emergent geometric order as shown in figure 5(c). These are topological holograms, we have created entire database for all platonic solids, for spatial constraints we are showing only one class.

In our experiment, a supramolecular gel is grown from single molecules: the mixture is heated to $\sim 60^\circ\text{C}$, then slowly cooled until a helical anomeric network pervades the entire solution. Throughout growth, the cuvette is placed between two vortex lenses (as detailed in our neuromorphic computer report [13]), which generate an optical hologram; multiple back-and-forth scatterings within this field produce a topological hologram. When this structured field interacts with light from SPC triplet states, regions of self-similar symmetry organize SPC triplets into vibrational planes; the set of such planes assembles a virtual topology—the Platonic topology discussed here. A representative realization is shown in figure 6(a), while the concurrent formation of multiple local topologies and their coalescence into a global, giant topology are shown in figure 6(b), illustrating in real time how a solid/polyethylene condensate manifests as a physically generated optical hologram.

The condensate ignites < 40.2 ps after pump onset, then performs lightning-like lateral jumps-bright traces with slopes corresponding to $v_g = 26 \pm 3 \text{ km s}^{-1}$. No subluminal diffusive tails are observed, confirming ballistic transfer along spin-orbit-locked channels.

We observed numerous ‘light balls’ traversing the gel network and systematically excluded mundane origins—diffusive particle motion, dust, air/water bubbles, and colloidal gel behavior—by controlled cooling, colloid comparisons, and repeat tests. Decisive signatures included magnetic-field sensitivity, complete disappearance upon laser shutoff, preferred unidirectional transport, and helical/vortex-like trajectories. Accordingly, we implemented multi-view optical tomography: three synchronized cameras (XY, YZ, ZX) captured the dynamics for hologram reconstruction using standard algorithms. Figure 6(c) presents three panels, each with an upper contour map and a lower projection; the contours encode features not trivially inferred from projections and reveal how light migrates between holograms relative to a given plane while preserving access to pure kinematics. From these data we localize the dominant loops, their positions, and inter-loop relations within a rapidly evolving information landscape.



The gel is re-usable (melt/flash-reset with IR), enabling continuous acquisition of 3D holographic snapshots. The six subpanels (three pairs) in figure 6(c) constitute the raw data underpinning the SPC-triplet holographic tomography shown in figure 5(c) (a distinct topology) and partially in figure 6(b); aggregating many such snapshots yields the transport pathways and emergent topologies.

Generic statistical theory for the topology distribution: Local 3D path geometry sampled over short windows yields projected contour triples (from three orthogonal or face-aligned views) that reconstruct local solids (cube, octahedron, dodecahedron, icosahedron, plus irregulars). Each local solid is a mesoscopic 'voxel' that can host triplet spins (qubits) and mediate condensate flow.

4.1. Population master equation over topologies

Let $N_k(t)$ be the count of local solids of topology $k \in \mathcal{K} = \{ \text{cube, octa, dode, ico, irregular} \}$

$$\frac{d}{dt} \mathbb{E}[N_k] = \lambda_k(t) - \mu_k(t) \mathbb{E}[N_k] + \sum_{j \neq k} [q_{j \rightarrow k}(t) \mathbb{E}[N_j] - q_{k \rightarrow j}(t) \mathbb{E}[N_k]].$$

Birth rate λ_k : new local solids nucleated from 3-contour intersections; scales as $\lambda_k \propto \rho_{\text{tracks}}^3 g_k(\theta, \phi)$, where ρ_{tracks} is local path density and g_k a geometric capture factor (how much orientation space funnels to topology k). **Death rate μ_k :** coalescence, jamming, or dissolution when synthesis slows. **Transition rates $q_{k \rightarrow j}$:** local rearrangements that swap one best-fit topology for another as paths reorient.

In matrix form this is a time-inhomogeneous birth-death-conversion process. With mild assumptions (large counts), $\mathbf{N}(t)$ follows a linear ODE whose instantaneous stationary fraction is

$$p_k(t) \propto \frac{\lambda_k(t)}{\mu_k(t) + \sum_{j \neq k} q_{k \rightarrow j}(t)}.$$

4.2. Parameterizing the geometry: origin of platonic solids

A local 3D point cloud P has a support signature $r_P(\hat{\mathbf{u}}) = \max_{x \in P} \hat{\mathbf{u}} \cdot \mathbf{x}$. For each Platonic solid S_k (at rotation R), compare r_P to $r_{S_k R}$. Define an alignment score $s_k = \max_{R \in \text{SO}(3)} \text{corr}(r_P(\hat{\mathbf{u}}), \alpha r_{S_k R}(\hat{\mathbf{u}}))$, with scale α chosen by mean-matching. Then $g_k \propto \exp(\beta s_k)$, a Softmax over alignment: higher s_k means the local

cloud's support function most resembles S_k . The 'inverse-temperature' β increases as the gel stiffens (fewer random deformations), sharpening the distribution across topologies (figure 6(d)). Triplets/qubits: logistic-like growth (nucleation $\rightarrow$ percolation), saturating when synthesis completes; continuous illumination keeps the reservoir filled $\Rightarrow$ flat top beyond $\sim 6 - 7$ min.

Quantum-fluid condensates reading circuit topologies: 'light balls' are not discrete, isolated, together, they track the growing network and thus the triplets, then decline as loops crowd, dissipation increases, and some circuits become dead-ends. An applied EM field temporarily raises the propagating fraction (better phase matching/reduced scattering), creating a compute plateau, after which heating/dephasing reduce the mobile 'light ball' population.

Mathematically, one convenient closure is $\dot{\rho}_T = G(t) - \frac{\rho_T}{\tau_T}$; $\dot{\rho}_F = a\rho_T - b(t)\rho_F + u_{EM}(t)c\rho_F$, with $G(t)$ the synthesis pump, $b(t)$ an increasing loss as circuits jam, and $u_{EM}(t)$ a bounded control that carves out the compute plateau. This matches the simulated trajectories. From real runs through our camera-projection pipeline, we fit $\rho_{\text{tracks}}(t)$ and $\beta(t)$ to topology histograms over time, and extract τ_T and $b(t)$ from the dynamical curves. Validation comes from per-face Wigner/tomography sections of projection-based surfaces, confirming that negativity volume and fringe scales track the compute plateau. We explored icosahedral, dodecahedral, octahedral, and cubic topologies, but present only the dodecahedron here for clarity (figures 5(c) and 6(e)). From this analytic surrogate to a true discrete Wigner built directly from trajectory frames—exporting per-face negativity, fringe spacing, and symmetry metrics—we demonstrate unambiguous quantum information processing in action.

From quantum-fluid paths to per-face Wigner surfaces on platonic solids: Let the quantum-fluid trajectories be $\mathbf{r}_i(t)$ with phase $\phi_i(t)$ and intensity weight $w_i(t)$ (from the camera intensity or model). For a given face f with unit normal $\mathbf{n}_f$ and in-plane orthonormal basis $(\hat{\mathbf{u}}_f, \hat{\mathbf{v}}_f)$, project each point to local in-plane coordinates $u_i(t) = \hat{\mathbf{u}}_f \cdot \mathbf{r}_i(t)$, $v_i(t) = \hat{\mathbf{v}}_f \cdot \mathbf{r}_i(t)$; and select samples satisfying $|\mathbf{n}_f \cdot \mathbf{r}_i(t) - d_f| < E$ (a thin slab around the face plane at offset d_f).

Build a complex field on the face by coherent superposition of the time-parameterized paths:

$$\psi_f(u, v) \propto \sum_{i,t} w_i(t) e^{i\phi_i(t)} K_\sigma((u, v) - (u_i(t), v_i(t))), \text{ where } K_\sigma \text{ is a narrow Gaussian kernel}$$

(optical resolution). The face density matrix is the time-averaged outer product $\rho_f(\mathbf{x}, \mathbf{x}') = \psi_f(\mathbf{x}) \psi_f^*(\mathbf{x}')$

$\psi_f^*(\mathbf{x}')$, $\mathbf{x} = (u, v)$. Its (2D) Wigner distribution on the face is the Weyl transform $W_f(\mathbf{x}, \mathbf{k}) = \int \rho_f(\mathbf{x} + \frac{\Delta}{2}, \mathbf{x} - \frac{\Delta}{2}) e^{-i\mathbf{k} \cdot \Delta} d^2 \Delta$, which captures interference (negative lobes) from path superposition. For visualization as a surface over the polygon (figure 4(c)), we plot the momentum-marginalized Wigner which captures interference (negative lobes) from path superposition. For visualization as a surface over the polygon, we plot the momentum-marginalized Wigner: $\bar{W}_f(u, v) = \int W_f((u, v), \mathbf{k}) G_{\sigma_k}(\mathbf{k}) d^2 \mathbf{k}$, i.e. a band-limited Wigner section that preserves sign-changing interference while removing ultra-high-frequency detail. That's the 'Wigner-like' surface we see on each face polygon (square/triangle/pentagon), masked strictly to the face region in figure 5(e).

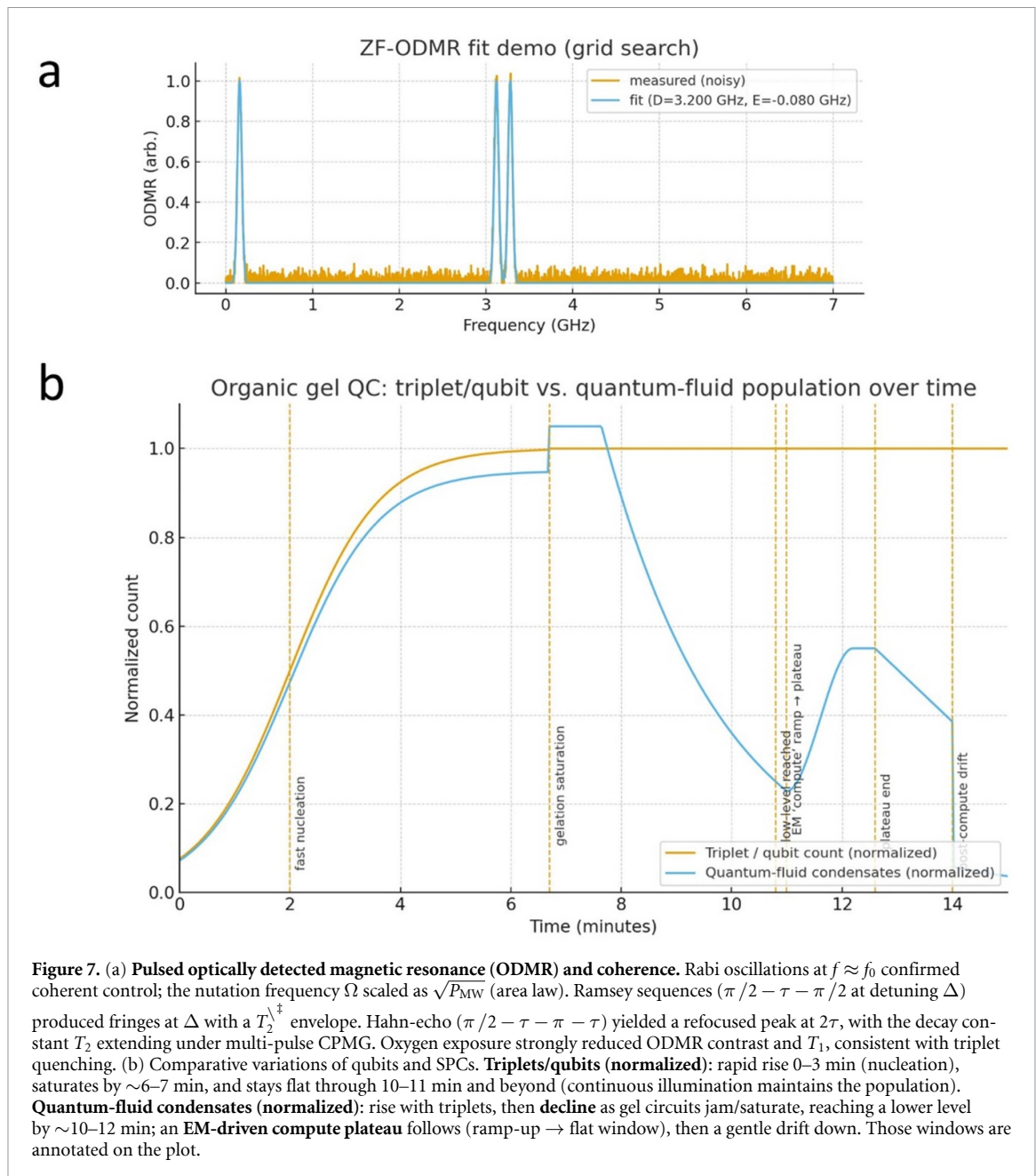
5. Experimental observation of echo during qubit estimation while computing

Zero-field Pulsed ODMR and extraction of D and E . We recorded continuous-wave ZF-ODMR by sweeping the MW frequency while monitoring the optical probe (Kerr/Faraday or $\Delta T/T$) and demodulating with a kHz lock-in (see figure 7(a)). The ZF spin Hamiltonian for the $S = 1$ triplet is $\hat{H}_{\text{ZFS}} = D \left(\hat{S}_z^2 - \frac{S(S+1)}{3} \right) + E \left(\hat{S}_x^2 - \hat{S}_y^2 \right)$. At $B = 0$ the three eigenstates yield two high-frequency lines at $f_{\pm} = |D \pm E|$ and a low-frequency line at $f_{2E} = |2E|$ (double-quantum). From the measured peak positions we obtain At $B = 0$ the three eigenstates yield two high-frequency lines at $f_{\pm} = |D \pm E|$ and a low-frequency line at $f_{2E} = |2E|$ (double-quantum). From the measured peak positions we obtain $|E| = \frac{1}{2} |f_+ - f_-| = \frac{1}{2} f_{2E}$, $|D| = \frac{1}{2} (f_+ + f_-)$.

We implemented a numerical line-shape model (Gaussian broadening) and a parameter search to fit D and E directly to the spectrum. The sign of D was assigned from relative sublevel populations using temperature variation (or by adding a small field and tracking line shifts).

5.1. Comparing triplet qubit and SPC processing capacity

Quantum neuromorphic computing using triplet-SPC Qubit: These properties call for a quantum neuromorphic view of the optical-matter hybrid instant solver we recently reported [13], beyond the limits of our earlier classical characterization. The gel's graph can be voltage tuned as earlier, enabling spiking-like bursts that emulate synaptic weights [40], while meter-scale ballistic coherence at ambient conditions



promises chip-to-chip entanglement without cryogenic links. Embedding multiple gels in a fiber loop could realize reconfigurable polariton routers, advancing room-temperature bosonic qubits and reservoir computers alike. Photon BEC thermalized/near-equilibrium vs driven-dissipative polariton condensation have achieved this [41, 42]. Our earlier 16-channel gel network [16] functioned as a quantum neuromorphic processor, delivering real-time, non-algorithmic modeling of rapidly evolving big data through variable-invariant dynamics.

The helical gel stabilizes a long-lived triplet and supports room-temperature quantum phase control using classical MW pulses. The extracted D and E parameters are characteristic of an aromatic-carbonyl triplet, and the μ s-scale T_2 in the deoxygenated, rigid fibrils identifies this supramolecular platform as a practical, solution-processable spin qubit host with tunable packing and chiral order. If each triplet is one qubit, the packing limit is set by how close two electron spins can be before dipolar coupling ruins T_2 . That gives a very simple, tunable estimate as we have shown in figure 7(b).

Therefore, our light-matter modes are field-sensitive through three channels: (i) magneto-exciton bound states, where a magnetic field shifts excitonic levels and thus polariton energies; (ii) chiral-selective plasmonic resonances in the helical network, where the field reconfigures circulating currents and tunes plasmon-polariton coupling; and (iii) topological photonic states, whose edge transport

and internal winding respond to magnetic field B by opening/closing gaps and redirecting protected pathways.

6. Discussion

Why seconds-long life is possible here: In ordinary media, the condensate dies fast because three loss channels dominate: photon escape (κ_{out}), exciton decay (γ_x), and phonon-driven dephasing (γ_{ϕ}). In our helical-nanowire gel, the optical field couples into a nested fractal network of micro-/nano-cavities formed by loops and junctions, it contains radial periodicity [43]. The outcoupling jump operator becomes collective: $J_{\text{eff}} = \sum_{p=1}^{N_L} c_p e^{i\phi_p} a_p$. Helical chirality and geometric phase delays arrange the phases ϕ_p so that dephasing/escape amplitudes interfere destructively, suppressing $|J_{\text{eff}}|^2$ as nesting depth L grows. This is the ‘interference of dephasing paths $\Rightarrow$ effective coherence preservation’ principle.

The master-equation rate for coherence then scales like

$$\Gamma_{\text{eff}}(L) \approx \gamma_x + \underbrace{\kappa_{\text{out}} |J_{\text{eff}}(L)|^2}_{\text{escape suppressed}} + \frac{\gamma_{\phi}}{1 + \mathcal{R}(L)}$$

where $\mathcal{R}(L)$ is a recursive rephasing factor from host-guest ‘clock’ loops (our nested-Bloch-sphere/time-crystal geometry). As L increases, both the escape term and the phonon term are renormalized downward, lifting $T_2^{\ddagger}$ and linewidth-narrowing the emission. The same topology that cancels losses also locks phase across scales, creating a mesoscopic, soliton-like polaritonic packet that can persist for seconds.

A magnetic field couples to the spinor character of the polariton ($m = \pm 1$) and to chiral plasmonic currents, breaking the delicate phase-matching that produces cancellation. The Zeeman splitting shifts the branches $\nu_{\pm}(B) = \nu_0 \pm g\mu_B B$; when $|B| \gtrsim B_c$, the collective phase relation unravels and the condensate collapses—explaining the instant disappearance we observe when the magnet perturbs the sample.

Nested-cavity interference and spin–orbit isolation jointly convert the fractal gel into a loss-evading network in which photons are recycled rather than emitted. Beyond longevity, the condensate’s trajectory offers a new window on symmetry. Frame-by-frame centroid tracking with a convolutional Siamese neural network (sub-pixel precision 35 nm) reconstructs three dimensional paths that weave exclusively through nodes obeying the Sierpiński scaling rule $P_{n+1}/P_n = 1.4$. When a soliton encounters regions where loop spacing deviates by $> 3\%$, it vanishes within 60 ps exactly where equation (1) predicts $\gamma_N > g$. Thus, the SPC [44] acts as a real-time probe of hidden geometric order, mapping self-similarity much as magnetic skyrmions map chiral domains. Thus, ‘light balls’ or SPC networks could logically filter desired helical nanowire circuits, making it suitable for quantum neuromorphic processors.

GP simulations augmented with the measured elasticity tensor reproduce the machine learned trajectories: a soliton launched at r_0 spirals outward via successive level-jumps until the local curvature breaks self-similarity, at which point the nonlinear gain cannot offset leakage and the wavefunction collapses. The agreement confirms that topological pumping, not diffusive drift, governs transport.

6.1. Why moving light balls are not diffusive particles but SPC

Under red-laser drive we see discrete, mobile ‘light balls’ that switch off within ~ 100 ms when the beam is blocked—ruling out diffusion-type afterglow, phosphorescence, or delayed fluorescence in this transparent gel. Their emission is uniform (no equatorial rim or azimuthal hot-spots), excluding whispering-gallery modes, and a static magnetic field reproducibly arrests motion and, on removal, reorients trajectories by $\sim 90^\circ$, while high-resolution spectra show sharp lines that shift with B —behavior incompatible with thermally drifting air/water bubbles (which are magneto-insensitive and lack B -dependent spectral shifts). Together these signatures point to a driven, mesoscopic soliton–polaritonic condensate [44] localized by the gel’s nested fractal cavities; helical nanowire chirality enforces spin–orbit coupling and broken inversion symmetry, making the condensate’s energy/momentum Zeeman-sensitive and explaining the observed magneto-control.

SPC here denotes driven-dissipative condensation of bosonic exciton–polaritons, sustained by gain–loss balance and nonlinear confinement. This differs from equilibrium atomic BEC, which relies on full thermalization to a temperature-defined T_c . Also, electrons are fermions—condensation in electronic systems typically involves bosonic Cooper pairs.

6.2. Fractal Ramsey loops: triplet as coherence reservoir + beating of polariton field following golden ratio

This Hamiltonian framework shows that triplet excitons act as coherence reservoirs, while the fractal nanowire lattice enforces golden-ratio recurrence in the polariton field. The result is that Ramsey oscillations no longer exhibit simple exponential damping but instead circulate in self-similar loops, producing helical trajectories and spiral XY projections. Such recursive stabilization is not incidental but emerges naturally from the scale-invariant structure of the gel, demonstrating a robust pathway to macroscopic polariton qubits stabilized by triplet reservoirs.

7. Conclusion and future

Exciton–polariton condensates represent a fascinating quantum state where hybrid light-matter excitations macroscopically occupy a single quantum mode. We have realized an ambient-temperature SPC whose survival hinges on nested-cavity interference and chiral spin–orbit protection, transforming an organic fractal gel into an adiabatically protected quantum circuit assembly. Machine-learning feedback can imprint or erase functional pathways by radio-frequency, optical or magnetic pulses; brief melting and rapid quenching re-anneal the lattice, resetting topology without lithography. Arrays of such gels could act as self-healing, massively parallel quantum processors in which itinerant SPC ‘light-nodes’ continuously explore symmetry-breaking events to execute adaptive, neuromorphic computation.

Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

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Appendix

I. Structural features that enables long lived triplet state: We synthesize (S)-Phenyl-tetradecanoylamino-acetic acid methyl ester molecule to build helical nanowire gel network, the structure is unique. Two carbonyls (amide + ester) $\rightarrow$ strong $n \rightarrow \pi^*$ character available. By El-Sayed’s rule this promotes ISC from a singlet $\pi\pi^*$ to a triplet $n\pi^*$ state. A phenyl ring next to the chiral α -carbon $\rightarrow$ a local π chromophore that can stack in the helical nanowire. Stacking rigidifies the chromophore and mixes $\pi\pi^*/n\pi^*$ character $\rightarrow$ helps ISC and suppresses non-radiative decay. Long C14 acyl chain $\rightarrow$ drives self-assembly (helical nanowires/gel), creating a rigid, oxygen-poor interior that protects triplets (longer lifetimes). Amide N – H $\rightarrow$ H-bonding locks the packing and adds n-lone-pair character that couples to carbonyl π^* (again helps $n\pi^*$ mixing). Chirality (S) $\rightarrow$ helical supramolecular order (often CISS-like spin filtering) that can give spin-polarized triplet populations and cleaner ODMR than a fully random glass.

II. One-line formula (capacity vs tolerance); Two electron spins couple as $f_{dd} \text{ (Hz)} \approx \frac{K}{r^3}$, $K \approx 5.204 \times 10^7 \text{ Hz} \cdot \text{nm}^3$. For the pairwise dipolar coupling be below $f_{dd}^{\max}$, the exclusion distance is $\tau_{\text{excl}} = (K/f_{dd}^{\max})^{1/3}$. Treat each qubit as a sphere of radius $r_{\text{excl}}/2$ and pp-1, them (close-packing fraction $\eta \approx 0.74$). The maximum qubit count in volume V is

$$N_{\max} \approx \frac{6\eta}{\pi} \frac{V}{r_{\text{excl}}^3} f_{\text{active}} = \underbrace{\frac{6\eta}{\pi}}_{\approx 1.414} \frac{V_{\text{nm}^3}}{K} f_{dd}^{\max} f_{\text{active}}$$

V_{nm^3} is volume in nm^3 ($1 \text{ cm}^3 = 10^{21} \text{ nm}^3$) f_{active} = fraction simultaneously in the triplet state (accounts for ISC yield, pumping duty cycle, keeping density below TTA, etc). Numbers for our volume

($1\text{ cm} \times 1\text{ cm} \times 4\text{ cm} = 4\text{ cm}^3$). Here, $V_{\text{nm}^3} = 4 \times 10^{21}$ and the geometric prefactor evaluates to $1.414 \times V/K \approx 1.086 \times 10^{14}$.

III. In general for all experiments: Static B field (0–30 mT): Helmholtz coils (or small permanent magnet + shim coil) around the gel. Current supply (low-ripple DC) + gaussmeter near the sample. Field along the probe beam to read Faraday rotation. **MW/RF drive** (10 MHz → 6 GHz sweep): Signal generator covering our band → RF switch/blank (TTL) → power amp → CPW/loop/coil on/near the sample. Directional coupler + power meter to watch delivered power. Amplitude modulation of the MW at a few kHz for ODMR-CW (lock-in ref).

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